

Sumcheck-free hypercube inner products from table-form Kronecker encodings

Research note (companion to Kronecker-FRI)

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Abstract

All protocols of this note are built on the hash-based commitment Kronecker-FRI and inherit its post-quantum security basis. We encode a multilinear polynomial by its *table*, $V_f(X) = \sum_w f(w)X^w$, instead of its coefficients. Two quantities then become single coefficients of univariate products: the evaluation $\tilde{f}(z)$, with a product-form kernel, and the hypercube inner product $\sum_w a(w)b(w)$ of two committed tables, through the reversal $V_b^*(X) = X^{N-1}V_b(1/X)$. Because a smooth domain is closed under inversion, the verifier reads V_b^* from the commitment to b as a virtual word, and the Kronecker-FRI machinery (opening polynomial, virtual word, one folding test) proves $\sum_w a(w)b(w) = S$ with no sumcheck, with soundness error $3M/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$. With geometric weights γ^w and DEEP quotients as further virtual words, the same method gives a sumcheck-free Hadamard check $a \circ b = c$ for three committed tables, with soundness error $2(N-1)/|\mathbb{F}| + 8M/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$. Finally, any list of evaluations, inner products and Hadamard checks on the same commitments is proved by one folding test, with soundness error $2(N-1)/|\mathbb{F}| + tM/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$ for $t+1$ batched words; the term $2(N-1)/|\mathbb{F}|$ is paid once, whatever the number of Hadamard checks. The results of Sections 1–4 are machine-checked in Lean 4. Section 5 reduces sparse matrix–vector products, and hence R1CS, to the same statements with lookups by logarithmic derivatives, so an R1CS argument needs no sumcheck at all; its soundness and completeness are machine-checked in Lean 4 end to end. Measured on the same Kronecker-FRI commitment and proof engine, a sumcheck-free inner product costs the same as a sumcheck followed by a batched opening, a Hadamard check 1.7 times more than a zerocheck, and the R1CS argument 19 times more than the core of Spartan, which proves 2^{15} constraints on Kronecker-FRI in 0.29s with 251 KiB proofs; the sumcheck-free arguments have a number of rounds independent of n and a verifier that stays succinct.

Setting. Notation and parameters are those of Kronecker-FRI (KF) [1]: $n \geq 1$, $N = 2^n$, a smooth domain $L \leq \mathbb{F}^\times$ of order $M = 2^{n+R}$ with $R \geq 2$, codes $\mathcal{C}_j$, the folding test $\Pi_{\text{FT}}[\ell, \kappa]$ with error $\varepsilon_{\text{fold}}$ (KF Thm 5.24), fibre distance Δ_0^{fib} , and $0 < \delta \leq \delta^* = (1 - \rho)/2$. A bit vector $w \in \{0, 1\}^n$ is identified with the integer $\sum_k w_k 2^{k-1} \in [0, N-1]$.

1 Table encoding and two extraction identities

Definition 1 (Table encoding, reversal). For a table $f : \{0, 1\}^n \rightarrow \mathbb{F}$, $V_f(X) = \sum_w f(w)X^w \in \mathbb{F}[X]_{<N}$ and $\text{Enc}^T(f) = \text{ev}_L(V_f)$. For $P \in \mathbb{F}[X]_{<N}$, its reversal is $P^*(X) = X^{N-1}P(1/X)$, so $[X^j]P^* = [X^{N-1-j}]P$ and $(P^*)^* = P$.

$f \mapsto V_f$ is a linear bijection onto $\mathbb{F}[X]_{<N}$, so Enc^T is a linear bijection onto $\mathcal{C}_0$ (as KF Lemma 6.3(1)); it is the Kronecker encoding of the multilinear polynomial whose coefficients are the values of f , and needs no Möbius transform. In particular $\text{Enc}^T(f) = \text{Enc}(f)$ as maps on vectors

of length N : a table-form commitment is an ordinary Kronecker-FRI commitment to the vector of values, with the same Merkle tree, and all of KF's results on commitments (unique decoding, extraction, batching) apply to it unchanged.

Lemma 2 (Evaluation). *For $z \in \mathbb{F}^n$ let $E_z^*(X) = \prod_{k=1}^n (z_k + (1 - z_k)X^{2^{k-1}})$. Then $\tilde{f}(z) = [X^{N-1}] V_f E_z^*$, where $\tilde{f}$ is the multilinear extension of f .*

Proof. Expanding the product, $E_z(X) := \sum_w \text{eq}(w, z) X^w = \prod_k ((1 - z_k) + z_k X^{2^{k-1}})$, and $X^{N-1} E_z(1/X) = \prod_k X^{2^{k-1}} ((1 - z_k) + z_k X^{-2^{k-1}}) = E_z^*(X)$ since $N - 1 = \sum_k 2^{k-1}$. Hence $[X^{N-1-w}] E_z^* = [X^w] E_z = \text{eq}(w, z)$ and $[X^{N-1}] V_f E_z^* = \sum_w f(w) \text{eq}(w, z) = \tilde{f}(z)$. $\square$

Lemma 3 (Inner product). *For tables a, b : $\sum_w a(w)b(w) = [X^{N-1}] V_a V_b^*$.*

Proof. $[X^{N-1}] V_a V_b^* = \sum_{i=0}^{N-1} [X^i] V_a [X^{N-1-i}] V_b^* = \sum_i [X^i] V_a [X^i] V_b = \sum_w a(w)b(w)$. $\square$

Both are instances of one fact: for $U \in \mathbb{F}[X]_{<N}$ and a public $K \in \mathbb{F}[X]_{<N}$, $[X^{N-1}] UK = \sum_i [X^i] U [X^{N-1-i}] K$. KF's split lemma (Lemma 4.10) and identity lemma (Lemma 4.12) use only $\deg U, \deg K \leq N - 1$, so they hold verbatim for UK replaced by $V_a E_z^*$ or $V_a V_b^*$:

Lemma 4 (Split and identity, general form). *Let $U, K, A, H \in \mathbb{F}[X]_{<N}$ and $v \in \mathbb{F}$. (1) There are $A, H \in \mathbb{F}[X]_{<N}$ with $XUK = A + vX^N + X^{N+1}H$ iff $v = [X^{N-1}] UK$. (2) $\Phi = XUK - A - vX^N - X^{N+1}H$ has degree at most $2N$, and $\Phi = 0$ implies $v = [X^{N-1}] UK$.*

Proof. As in KF Lemmas 4.10 and 4.12: $\deg(UK) \leq 2N - 2$, and the three parts of the right-hand side occupy the disjoint degree ranges $[0, N - 1]$, $\{N\}$, $[N + 1, 2N]$; compare the coefficients of X^N . $\square$

Lemma 5 (Reversed words). *For $w \in \mathbb{F}^L$ put $w^*(\xi) = \xi^{N-1} w(\xi^{-1})$. Then: (1) $w = \text{ev}_L(P)$ with $P \in \mathbb{F}[X]_{<N}$ implies $w^* = \text{ev}_L(P^*)$; in particular $w \mapsto w^*$ maps $\mathcal{C}_0$ onto $\mathcal{C}_0$; (2) $w \mapsto w^*$ is a linear bijection of $\mathbb{F}^L$, and $w^* = c^*$ at ξ iff $w = c$ at ξ^{-1} ; (3) a set $T \subseteq L$ is fibre-closed iff $T^{-1} = \{\xi^{-1} : \xi \in T\}$ is, and $|T^{-1}| = |T|$; (4) $h(\xi) = w^*(\xi)$ is computed from the value of w at $\xi^{-1} \in L$, and a fold check at $\pm\xi$ reads w on the fibre $\{\pm\xi^{-1}\}$.*

Proof. L is a group, so $\xi \mapsto \xi^{-1}$ is a bijection of L mapping $-\xi$ to $-\xi^{-1}$ (as $-1 \in L$); and $\xi^{N-1} \neq 0$. (1) $\xi^{N-1} P(\xi^{-1}) = P^*(\xi)$, and $P \mapsto P^*$ is a bijection of $\mathbb{F}[X]_{<N}$. (2)–(4) follow from the first sentence. $\square$

2 The inner-product protocol

Definition 6 (Π_{IP}). Instance: oracles $w_a, w_b \in \mathbb{F}^L$ and $S \in \mathbb{F}$; honestly $w_a = \text{Enc}^T(a)$, $w_b = \text{Enc}^T(b)$ and $S = \sum_w a(w)b(w)$.

1. The prover sends $w_A = \text{ev}_L(A)$, where $XV_a V_b^* = A + SX^N + X^{N+1}H$ (Lemma 4); the verifier sends $\beta \in \mathbb{F}$.
2. Put the virtual words w_b^* (Lemma 5) and $h(\xi) = (\xi w_a(\xi) w_b^*(\xi) - w_A(\xi) - S\xi^N)/\xi^{N+1}$, and

$$w_0 = w_a + \beta w_b^* + \beta^2 w_A + \beta^3 h.$$

3. Run $\Pi_{\text{FT}}[\ell, \kappa]$ (or Π_{FT}^C , KF Prop. 5.28) on w_0 .

Per query point the verifier reads $w_a(\pm\xi)$, $w_b(\pm\xi^{-1})$ and $w_A(\pm\xi)$, and does $O(1)$ extra field operations beyond KF (no kernel to evaluate).

Theorem 7 (Completeness). *If $w_a = \text{ev}_L(V_a)$, $w_b = \text{ev}_L(V_b)$ and $S = \sum_w a(w)b(w)$, the honest prover is accepted with probability 1.*

Proof. By Lemma 3 and Lemma 4(1), $A, H \in \mathbb{F}[X]_{<N}$ exist; by Lemma 5(1) $w_b^* = \text{ev}_L(V_b^*)$; evaluating the identity at ξ gives $h = \text{ev}_L(H)$. So $w_0 = \text{ev}_L(V_a + \beta V_b^* + \beta^2 A + \beta^3 H)$ with a polynomial of degree $< N$, and KF Lemma 5.9(1) applies. $\square$

Definition 8 (Relation). $\mathcal{R}_{\text{IP}}^\delta$: $((w_a, w_b; S), (a, b))$ with $\Delta_0^{\text{fib}}(w_a, \text{Enc}^T(a)) \leq \delta$, $\Delta_0^{\text{fib}}(w_b, \text{Enc}^T(b)) \leq \delta$ and $\sum_w a(w)b(w) = S$.

Lemma 9 (Batching, degree 3). *Let $\mathcal{G}$ be the set of β for which w_0 agrees with some $c \in \mathcal{C}_0$ on a fibre-closed $T \subseteq L$ with $|T| \geq (1 - \delta)M$. If $|\mathcal{G}| > 3M$, then $(w_a, w_b; S)$ has a witness in $\mathcal{R}_{\text{IP}}^\delta$.*

Proof. The words w_0 form the curve of degree 3 generated by (w_a, w_b^*, w_A, h) , and every $\beta \in \mathcal{G}$ has $\Delta(w_0, \mathcal{C}_0) \leq \delta$. By correlated agreement for curves (KF Thm 2.16, from [2], with $t = 3$) there are $P_0, P_1, P_2, P_3 \in \mathbb{F}[X]_{<N}$ and a set Agr , $|\text{Agr}| \geq (1 - \delta)M$, on which $w_a = \text{ev}(P_0)$, $w_b^* = \text{ev}(P_1)$, $w_A = \text{ev}(P_2)$, $h = \text{ev}(P_3)$ (take Agr maximal).

Value. Let $\Phi = XP_0P_1 - P_2 - SX^N - X^{N+1}P_3$. On Agr , $\Phi(\xi) = \xi w_a(\xi)w_b^*(\xi) - w_A(\xi) - S\xi^N - \xi^{N+1}h(\xi) = 0$ by the definition of h . Since $\deg \Phi \leq 2N < (1 - \delta)M \leq |\text{Agr}|$ (KF Lemma 7.2, which uses $R \geq 2$), $\Phi = 0$, and Lemma 4(2) gives $S = [X^{N-1}]P_0P_1$. Let a, b be the tables with $V_a = P_0$ and $V_b = P_1^*$; then $V_b^* = P_1$ and $S = \sum_w a(w)b(w)$ by Lemma 3.

Bad challenges. For $\xi \notin \text{Agr}$, $d_\xi(Z) = \sum_{i=0}^3 Z^i(u_i - \text{ev}(P_i))(\xi)$, with $(u_0, \dots, u_3) = (w_a, w_b^*, w_A, h)$, is a nonzero polynomial of degree ≤ 3 ; its roots over all such ξ form a set Bad with $|\text{Bad}| \leq 3\delta M < 3M < |\mathcal{G}|$. Take $\beta \in \mathcal{G} \setminus \text{Bad}$ with its c and T , and $\hat{c} = \sum_i \beta^i \text{ev}(P_i) \in \mathcal{C}_0$. Then c and $\hat{c}$ agree with w_0 on $T \cap \text{Agr}$, of size $\geq (1 - 2\delta)M \geq N$, so $c = \hat{c}$; and for $\xi \in T$, $d_\xi(\beta) = w_0(\xi) - \hat{c}(\xi) = 0$ forces $\xi \in \text{Agr}$. So $T \subseteq \text{Agr}$.

Distances. $w_a = \text{ev}(V_a)$ on the fibre-closed T , so $\Delta_0^{\text{fib}}(w_a, \text{Enc}^T(a)) \leq \delta$ (KF Lemma 5.11(2)). $w_b^* = \text{ev}(V_b^*) = \text{ev}(V_b)^*$ on T , so by Lemma 5(2) $w_b = \text{ev}(V_b)$ on T^{-1} , which is fibre-closed of the same size (Lemma 5(3)); so $\Delta_0^{\text{fib}}(w_b, \text{Enc}^T(b)) \leq \delta$. $\square$

Theorem 10 (Soundness). *If $(w_a, w_b; S)$ has no witness in $\mathcal{R}_{\text{IP}}^\delta$, every prover is accepted with probability at most*

$$\frac{3M}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa < \frac{4M}{|\mathbb{F}|} + (1 - \delta)^\kappa.$$

Proof. Identical to KF Thm 7.8, with Lemma 9 in place of KF Lemma 7.4: for $\beta \notin \mathcal{G}$ (probability $\geq 1 - 3M/|\mathbb{F}|$) and good r (probability $\geq 1 - \varepsilon_{\text{fold}}$, KF Thm 5.24(1)), the codeword chain (KF Prop. 5.23) shows that $\text{dens}_\ell(w_\ell) \geq 1 - \delta$ would put β in $\mathcal{G}$, the witness set $\mathcal{W}_\ell(w_\ell)$ being fibre-closed (KF Lemma 7.7); so the query term is at most $(1 - \delta)^\kappa$ (KF Lemma 3.3). $\square$

Round-by-round knowledge soundness, the extractor (decode w_a and w_b at radius δ^*) and binding follow as in KF Thm 7.11 and Cor. 7.12, with round-1 error $3M/|\mathbb{F}|$.

Remark 11 (One commitment, three kinds of openings). The same table-form commitment $\text{Enc}^T(f)$ supports: evaluation openings at any z (KF's protocol with the kernel E_z^* of Lemma 2, evaluated by the verifier with $O(n)$ operations); hypercube sums ($z = (\frac{1}{2}, \dots, \frac{1}{2})$ up to 2^n , or the kernel $\prod_k (1 + X^{2^{k-1}})$ directly); and inner products with any other committed table (Π_{IP}). Inner products with a *public* table p are the special case $K = V_p^*$, useful when $V_p^*(\xi)$ is cheap to evaluate (product form). All of these batch with KF's batched openings (KF Thm 11.2).

3 A sumcheck-free Hadamard check

SNARKs of Spartan/HyperPlonk type need, beyond linear functionals and inner products, the statement $a(w)b(w) = c(w)$ for all w (a *Hadamard check*), for three committed tables.

Geometric weights. For $\gamma \in \mathbb{F}$,

$$\sum_w \gamma^w a(w)b(w) = [X^{N-1}] V_a(\gamma X) V_b^*(X), \quad \sum_w \gamma^w c(w) = V_c(\gamma),$$

the first by Lemma 3 applied to the tables $w \mapsto \gamma^w a(w)$ and b . If $d = a \circ b - c \neq 0$, then $D_d(X) = \sum_w d(w)X^w$ is a nonzero polynomial of degree $\leq N-1$, and the two sides differ unless $D_d(\gamma) = 0$. The values $V_a(\gamma\xi)$ and $V_c(\gamma)$ lie outside L ; the prover commits to $Q(X) = V_a(\gamma X)$, and both are tied to the commitments by DEEP quotients, which are virtual words like h .

Definition 12 (Quotient words). For $w \in \mathbb{F}^L$ and $y, x \in \mathbb{F}$, put $q[w, y, x](\xi) = (w(\xi) - y)/(\xi - x)$, with the value 0 at $\xi = x$ if $x \in L$.

Lemma 13 (DEEP step on an agreement set). *Let $x, y \in \mathbb{F}$, $P, R \in \mathbb{F}[X]_{<N}$ and $T \subseteq L$ with $|T| > N+1$. If $w = \text{ev}(P)$ and $q[w, y, x] = \text{ev}(R)$ on T , then $P(x) = y$ and $R = (P - y)/(X - x)$. Conversely, if $x \notin L$, $w = \text{ev}_L(P)$ and $P(x) = y$, then $q[w, y, x] = \text{ev}_L((P - y)/(X - x))$, a polynomial of degree $\leq N-2$.*

Proof. $(X - x)R - P + y$ has degree $\leq N$ and vanishes on $T \setminus \{x\}$, which has more than N points, so it is 0; evaluate at x . The converse is the factor theorem. $\square$

Definition 14 (Π_{Had}). Instance: oracles $w_a, w_b, w_c \in \mathbb{F}^L$; honestly $w_f = \text{Enc}^T(f)$ for $f \in \{a, b, c\}$ and $a \circ b = c$.

1. The verifier sends γ , uniform in $\mathbb{F}$. The prover sends $w_Q = \text{ev}_L(Q)$, where $Q(X) = V_a(\gamma X)$.
2. The verifier sends θ , uniform in $\mathbb{F}$. The prover sends $y_1 = V_a(\gamma\theta)$, $y_3 = V_c(\gamma)$ and $w_A = \text{ev}_L(A)$, where $XQV_b^* = A + y_3X^N + X^{N+1}H$ (Lemma 4).
3. The verifier sends β , uniform in $\mathbb{F}$. With $h(\xi) = (\xi w_Q(\xi) w_b^*(\xi) - w_A(\xi) - y_3\xi^N)/\xi^{N+1}$ and

$$(u_0, \dots, u_8) = (w_a, w_c, w_Q, w_b^*, w_A, h, q[w_a, y_1, \gamma\theta], q[w_Q, y_1, \theta], q[w_c, y_3, \gamma]),$$

$$\text{put } w_0 = \sum_{i=0}^8 \beta^i u_i.$$

4. Run $\Pi_{\text{FT}}[\ell, \kappa]$ (or Π_{FT}^C) on w_0 .

The challenges are unrestricted: soundness holds for all values, and a pole $\gamma\theta, \theta, \gamma$ in L affects only completeness. Using y_1 for both $V_a(\gamma\theta)$ and $Q(\theta)$ is the consistency check $V_a(\gamma\theta) = Q(\theta)$. Per query point the verifier reads w_a, w_c, w_Q, w_A on the coset of ξ and w_b on the coset of ξ^{-1} , and does $O(1)$ field operations and three inversions per point (one batched inversion per query) beyond the folding test.

Theorem 15 (Completeness). *If $w_f = \text{Enc}^T(f)$ for $f \in \{a, b, c\}$ and $a \circ b = c$, the honest prover is accepted whenever $\gamma\theta, \theta, \gamma \notin L$, hence with probability at least $1 - 3M/|\mathbb{F}|$.*

Proof. $Q \in \mathbb{F}[X]_{<N}$; $y_1 = V_a(\gamma\theta) = Q(\theta)$; and $y_3 = V_c(\gamma) = \sum_w \gamma^w c(w) = \sum_w \gamma^w a(w)b(w) = [X^{N-1}] QV_b^*$, so A, H exist and $h = \text{ev}_L(H)$ as in Π_{IP} . By Lemma 13 the three quotient words are codewords when the poles lie outside L , and then w_0 is the codeword of a polynomial of degree $< N$. Each of the events $\gamma \in L$, $\theta \in L$, $\gamma\theta \in L$ has probability at most $M/|\mathbb{F}|$ (for $\gamma \neq 0$, at most M values of θ put $\gamma\theta$ in L ; for $\gamma = 0$, none). $\square$

Definition 16 (Relation). $\mathcal{R}_{\text{Had}}^\delta$: $((w_a, w_b, w_c), (a, b, c))$ with $\Delta_0^{\text{fib}}(w_f, \text{Enc}^T(f)) \leq \delta$ for $f \in \{a, b, c\}$ and $a \circ b = c$.

Lemma 17 (Batching, degree 8). Fix $\gamma, \theta \in \mathbb{F}$ and the prover's w_Q, y_1, y_3, w_A . Let $\mathcal{G}$ be the set of β for which w_0 agrees with some $c' \in \mathcal{C}_0$ on a fibre-closed $T \subseteq L$ with $|T| \geq (1 - \delta)M$. If $|\mathcal{G}| > 8M$, there are tables a, b, c and $\hat{Q} \in \mathbb{F}[X]_{<N}$ with

1. $\Delta_0^{\text{fib}}(w_f, \text{Enc}^T(f)) \leq \delta$ for $f \in \{a, b, c\}$, and $\Delta_0^{\text{fib}}(w_Q, \text{ev}_L(\hat{Q})) \leq \delta$;
2. $V_a(\gamma\theta) = y_1 = \hat{Q}(\theta)$;
3. $V_c(\gamma) = y_3 = [X^{N-1}] \hat{Q} V_b^*$.

Proof. Correlated agreement on one fibre-closed set (the step shared with Lemma 9; Lean: `curve_agreement`, $t = 8$) gives $P_0, \dots, P_8 \in \mathbb{F}[X]_{<N}$ and a fibre-closed T , $|T| \geq (1 - \delta)M > 2N \geq N + 2$ (KF Lemma 7.2), with $u_i = \text{ev}(P_i)$ on T . As in Lemma 9, $X P_2 P_3 - P_4 - y_3 X^N - X^{N+1} P_5$ vanishes on T and has degree $\leq 2N$, so $y_3 = [X^{N-1}] P_2 P_3$. Lemma 13 with $(P_0, P_6, \gamma\theta)$, (P_2, P_7, θ) and (P_1, P_8, γ) gives $P_0(\gamma\theta) = y_1$, $P_2(\theta) = y_1$ and $P_1(\gamma) = y_3$. Let $V_a = P_0$, $V_c = P_1$, $V_b = P_3^*$ and $\hat{Q} = P_2$; then (2) and (3) hold, and (1) follows from agreement on T (for w_b , on T^{-1} , Lemma 5). $\square$

Theorem 18 (Soundness). Let $R \geq 2$ and $0 < \delta \leq \delta^*$. If (w_a, w_b, w_c) has no witness in $\mathcal{R}_{\text{Had}}^\delta$, every prover is accepted with probability at most

$$\varepsilon_{\text{Had}} = \frac{2(N-1)}{|\mathbb{F}|} + \frac{8M}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa.$$

Proof. For $\delta \leq \delta^*$ each word has at most one codeword within fibre distance δ (KF Lemma 6.3(2)), so the tables and $\hat{Q}$ of Lemma 17 are the decodings of w_a, w_b, w_c, w_Q . It suffices to consider deterministic provers (a randomised prover is an average of deterministic ones), so w_Q is a function of γ and the round-2 messages are functions of (γ, θ) . Call (γ, θ) *good* if, for every choice of the prover's round-2 messages, $|\mathcal{G}| \leq 8M$. For good (γ, θ) the argument of KF Thm 7.8 applied to the rest of the protocol bounds the acceptance probability by $8M/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$. It remains to bound the probability that (γ, θ) is not good.

Case 1: some w_f has no table within fibre distance δ . Lemma 17(1) fails, so every (γ, θ) is good.

Case 2: w_a, w_b, w_c decode to $\hat{a}, \hat{b}, \hat{c}$ with $\hat{a} \circ \hat{b} \neq \hat{c}$. Let $D = D_{\hat{a} \circ \hat{b} - \hat{c}} \neq 0$. The event $D(\gamma) = 0$ has probability at most $(N-1)/|\mathbb{F}|$. Fix γ with $D(\gamma) \neq 0$, and the prover's w_Q (a function of γ). If w_Q has no codeword within fibre distance δ , (γ, θ) is good for every θ by Lemma 17(1). If it decodes to $\hat{Q} = V_{\hat{a}}(\gamma X)$, then Lemma 17(3) would give $\sum_w \gamma^w \hat{c}(w) = V_{\hat{c}}(\gamma) = [X^{N-1}] V_{\hat{a}}(\gamma X) V_{\hat{b}}^* = \sum_w \gamma^w \hat{a}(w) \hat{b}(w)$, i.e. $D(\gamma) = 0$; so every θ is good. Otherwise $\hat{Q} - V_{\hat{a}}(\gamma X)$ is a nonzero polynomial of degree $\leq N-1$, fixed before θ is drawn; the event $\hat{Q}(\theta) = V_{\hat{a}}(\gamma\theta)$ has probability at most $(N-1)/|\mathbb{F}|$, and outside it Lemma 17(2) fails, so θ is good. $\square$

So $\varepsilon_{\text{Had}} < (8M + 2N)/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$, against $3M/|\mathbb{F}|$ for Π_{IP} and $2M/|\mathbb{F}|$ for an evaluation. The curve has degree 8 because w_a and w_c must be in the batch themselves: a quotient word alone determines w_a on T only as a polynomial of degree $\leq N$, which does not identify the decoding.

Knowledge follows: if the acceptance probability exceeds ε_{Had} , the instance has a witness, and the extractor decodes w_a, w_b, w_c at radius δ (KF Cor. 7.12).

Remark 19 (Consequence). Linear functionals (evaluations, with the kernel E_z^*), inner products (Π_{IP}) and Hadamard checks (Π_{Had}) are all proved on table-form Kronecker-FRI commitments by one folding test each, with no sumcheck and a verifier that is an FRI verifier plus $O(1)$ operations per query point (plus $O(n)$ for E_z^*). For R1CS-type SNARKs what remains is the sparse matrix-vector products Az, Bz, Cz , which Spartan handles with its own sparse-polynomial commitment; whether those admit a coefficient-extraction form is the next question.

4 One folding test for a batch of statements

A SNARK needs many statements at once on the same commitments: Hadamard checks, inner products and evaluations. Since every statement of Sections 1–3 is checked by words in one curve, all of them can share one curve and one folding test.

Definition 20 (Statements). Let $w_1, \dots, w_r \in \mathbb{F}^L$ be commitments. A *statement* is one of the following, with indices in $\{1, \dots, r\}$:

- **Eval** (i, z, v) with $z \in \mathbb{F}^n$, $v \in \mathbb{F}$: the table f_i of w_i has $\tilde{f}_i(z) = v$;
- **IP** (i, i', S) with $S \in \mathbb{F}$: $\sum_w f_i(w) f_{i'}(w) = S$;
- **Had** (i, i', k) : $f_i \circ f_{i'} = f_k$.

For a list $\Sigma = (\sigma_1, \dots, \sigma_J)$ of $J \geq 1$ statements, let D be the set of commitment indices read directly (i in **Eval** and **IP**; i, k in **Had**), B the set of indices read reversed (i' in **IP** and **Had**), and $\mathcal{H} \subseteq \{1, \dots, J\}$ the positions of the Hadamard checks. A table may play several roles. The relation $\mathcal{R}_{\text{Batch}}^\delta$ holds for $((w_i)_i, \Sigma)$ and tables $(f_i)_{i \in D \cup B}$ if $\Delta_0^{\text{fib}}(w_i, \text{Enc}^T(f_i)) \leq \delta$ for all $i \in D \cup B$ and every statement of Σ holds for these tables.

For each statement σ_j the protocol uses a triple (U_j, K_j, v_j) , with $U_j, K_j \in \mathbb{F}[X]_{<N}$, and the identity $XU_jK_j = A_j + v_jX^N + X^{N+1}H_j$ of Lemma 4: (V_{f_i}, E_z^*, v) for **Eval** (i, z, v) (Lemma 2); $(V_{f_i}, V_{f_{i'}}^*, S)$ for **IP** (i, i', S) (Lemma 3); $(Q_j, V_{f_{i'}}^*, y_3^{(j)})$ with $Q_j(X) = V_{f_i}(\gamma X)$ for $\sigma_j = \text{Had}(i, i', k)$ (Section 3). The verifier reads $U_j(\xi)$ and $K_j(\xi)$ as $w_i(\xi)$ and $E_z^*(\xi)$, as $w_i(\xi)$ and $w_{i'}^*(\xi)$, or as $w_Q^{(j)}(\xi)$ and $w_{i'}^*(\xi)$.

Definition 21 (Π_{Batch}). Instance: $(w_i)_i$ and Σ .

1. The verifier sends γ , uniform in $\mathbb{F}$. For each $j \in \mathcal{H}$ the prover sends $w_Q^{(j)} = \text{ev}_L(Q_j)$.
2. The verifier sends θ , uniform in $\mathbb{F}$. For each $j \in \mathcal{H}$, with $\sigma_j = \text{Had}(i, i', k)$, the prover sends $y_1^{(j)} = V_{f_i}(\gamma\theta)$ and $y_3^{(j)} = V_{f_k}(\gamma)$; for each $j \leq J$ it sends $w_A^{(j)} = \text{ev}_L(A_j)$.
3. The verifier sends β , uniform in $\mathbb{F}$. The words are, in a fixed order: w_i for $i \in D$; w_i^* for $i \in B$; $w_Q^{(j)}$ for $j \in \mathcal{H}$; $w_A^{(j)}$ and $h_j(\xi) = (\xi U_j(\xi) K_j(\xi) - w_A^{(j)}(\xi) - v_j \xi^N) / \xi^{N+1}$ for $j \leq J$; and $q[w_i, y_1^{(j)}, \gamma\theta]$, $q[w_Q^{(j)}, y_1^{(j)}, \theta]$, $q[w_k, y_3^{(j)}, \gamma]$ for $j \in \mathcal{H}$, $\sigma_j = \text{Had}(i, i', k)$. They are $t+1 = |D| + |B| + 2J + 4|\mathcal{H}|$ words $u_0, \dots, u_t$; put $w_0 = \sum_{s=0}^t \beta^s u_s$.
4. Run $\Pi_{\text{FT}}[\ell, \kappa]$ (or Π_{FT}^C) on w_0 .

γ and θ are shared by all Hadamard checks. All $w_Q^{(j)}$ can be committed in one Merkle tree whose leaves hold one value of each, and likewise all $w_A^{(j)}$; a query then opens, beyond the folding test, one path per commitment in D , one per commitment in B (on the inverse coset), and at most two paths for the prover's oracles (one if $\mathcal{H} = \emptyset$), whatever J .

Theorem 22 (Completeness). *If $w_i = \text{Enc}^T(f_i)$ and every statement holds, the honest prover is accepted whenever $\mathcal{H} = \emptyset$ or $\gamma\theta, \theta, \gamma \notin L$; so with probability at least $1 - 3M/|\mathbb{F}|$.*

Proof. Each $w_A^{(j)}$, h_j and quotient word is a codeword, by the completeness proofs of Sections 2 and 3 and Lemma 2 for **Eval** (where E_z^* is a public polynomial of degree $< N$). $\square$

Lemma 23 (Batching, degree t). *Fix γ, θ and the prover's messages, and let $\mathcal{G}$ be the set of β for which w_0 agrees with some $c' \in \mathcal{C}_0$ on a fibre-closed $T \subseteq L$ with $|T| \geq (1 - \delta)M$. Let $R \geq 2$ and $0 < \delta \leq \delta^*$. If $|\mathcal{G}| > tM$, there are tables $(f_i)_{i \in D \cup B}$ and polynomials $\hat{Q}_j \in \mathbb{F}[X]_{<N}$ ($j \in \mathcal{H}$) such that $\Delta_0^{\text{fib}}(w_i, \text{Enc}^T(f_i)) \leq \delta$, $\Delta_0^{\text{fib}}(w_Q^{(j)}, \text{ev}_L(\hat{Q}_j)) \leq \delta$, and*

1. every **Eval** and **IP** statement of Σ holds for the f_i ;
2. for every $j \in \mathcal{H}$ with $\sigma_j = \text{Had}(i, i', k)$: $V_{f_i}(\gamma\theta) = y_1^{(j)} = \hat{Q}_j(\theta)$ and $V_{f_k}(\gamma) = y_3^{(j)} = [X^{N-1}] \hat{Q}_j V_{f_{i'}}^*$.

Proof. Correlated agreement on one fibre-closed set (**curve_agreement** with this t) gives polynomials of degree $< N$ for all the words and a fibre-closed T with $|T| \geq (1 - \delta)M > 2N$ on which all words agree with them. Tables: V_{f_i} is the polynomial of w_i for $i \in D$; for $i \in B$, V_{f_i} is the reversal of the polynomial of w_i^* , and w_i agrees with $\text{Enc}^T(f_i)$ on T^{-1} (Lemma 5). If $i \in D \cap B$, both give tables within fibre distance δ of w_i , which coincide by unique decoding. For each j , the identity of h_j on T and Lemma 4(2) give $v_j = [X^{N-1}] U_j K_j$ with the polynomials of the words; for **Eval** this is $v = \hat{f}_i(z)$ by Lemma 2, for **IP** it is $S = \sum_w f_i(w) f_{i'}(w)$ by Lemma 3, and for **Had** it is the second equality of (2) with $\hat{Q}_j$ the polynomial of $w_Q^{(j)}$. The three quotient words of each $j \in \mathcal{H}$ give the other equalities of (2) by Lemma 13, as in Lemma 17. $\square$

Theorem 24 (Soundness). *Let $R \geq 2$ and $0 < \delta \leq \delta^*$. If $((w_i)_i, \Sigma)$ has no witness in $\mathcal{R}_{\text{Batch}}^\delta$, every prover is accepted with probability at most*

$$\varepsilon_{\text{Batch}} = \frac{2(N-1)}{|\mathbb{F}|} + \frac{tM}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa.$$

Proof. As for Theorem 18: call (γ, θ) good if $|\mathcal{G}| \leq tM$ for all round-2 messages; for good (γ, θ) the acceptance probability is at most $tM/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$. If some w_i , $i \in D \cup B$, has no table within fibre distance δ , every pair is good by Lemma 23. Otherwise let $\hat{f}_i$ be the decodings; some statement σ_{j^*} is false for them, and j^* depends only on the instance. If σ_{j^*} is an **Eval** or **IP** statement, Lemma 23(1) fails, so every pair is good. If $\sigma_{j^*} = \text{Had}(i, i', k)$, the proof of Theorem 18 applies verbatim to this one check, with Lemma 23(2) for j^* in place of Lemma 17: the pair is good unless $D_{\hat{f}_i \circ \hat{f}_{i'} - \hat{f}_k}(\gamma) = 0$, or $\hat{Q}_{j^*} \neq V_{\hat{f}_i}(\gamma X)$ and $\hat{Q}_{j^*}(\theta) = V_{\hat{f}_i}(\gamma\theta)$, events of probability at most $(N-1)/|\mathbb{F}|$ each. $\square$

When $\mathcal{H} = \emptyset$ the same proof shows that every pair is good, so the first term can be omitted (this refinement is not in the Lean formalisation). Only one false statement is needed in the proof, so the term $2(N-1)/|\mathbb{F}|$ from γ, θ is paid once, not once per Hadamard check; the term $tM/|\mathbb{F}|$ grows linearly in the number of words, by $4M/|\mathbb{F}|$ per Hadamard check and $2M/|\mathbb{F}|$ per other statement. For $|\mathbb{F}| = p^2 \approx 2^{128}$, $M = 2^{22}$ and $t = 100$, this term is about 2^{-99} . Knowledge follows as before: if the acceptance probability exceeds $\varepsilon_{\text{Batch}}$, the instance has a witness, obtained by decoding each w_i .

n	one batched proof ($t + 1 = 16$)			four separate proofs		
	prove (ms)	verify (ms)	proof (KiB)	prove (ms)	verify (ms)	proof (KiB)
12	38.2	7.21	397.9	54.3	13.85	795.4
14	165.2	8.20	479.5	224.9	16.17	1017.2
16	716.2	8.90	587.5	959.8	18.70	1315.3
18	3154.6	10.22	701.2	4116.7	22.59	1658.7
20	13187.4	11.32	804.5	18012.6	25.43	1957.9

Table 1: The statements $\text{Had}(a, b, c)$, $\text{IP}(a, d, S)$, $\text{Eval}(a, z_1, v_1)$, $\text{Eval}(c, z_2, v_2)$ on four table-form commitments, proved by one Π_{Batch} proof or by four (each a Π_{Batch} proof of one statement). Configuration KF-8, 148 queries, 32-byte salts, $\mathbb{F}_{p^2}$; one core; medians. Source: `bench/batch.csv`, `cargo run --release --example batch_bench`. The prototype prover is not optimised.

One proof for the four statements is 2.0 to 2.4 times smaller and 1.9 to 2.2 times faster to verify than four proofs (2.4 and 2.2 at $n = 20$), and its prover takes 23 to 30% less time.

5 Sparse matrix–vector products without sumcheck

An R1CS instance asks for z with $(Az) \circ (Bz) = Cz$ for public sparse matrices A, B, C . Committing to z and to $a = Az$, $b = Bz$, $c = Cz$, the statement $a \circ b = c$ is a Hadamard check. This section proves $a = \mathbf{M}z$ for a public sparse matrix $\mathbf{M}$ (a *lincheck*) with the statements of Section 4 on affine forms of committed and public tables; lookups are handled with logarithmic derivatives (LogUp, Haböck [4]). Throughout, N is at least the length of z and the number of nonzero entries of each matrix, and all tables are padded to length $N = 2^n$; $R \geq 2$ and $0 < \delta \leq \delta^*$. Several matrices $\mathbf{M}_1, \dots, \mathbf{M}_J$ are handled together (for R1CS, $J = 3$), and $\text{char } \mathbb{F} > JN$.

5.1 Affine forms and public tables

A *public table* is a table P whose polynomial V_P (and hence V_P^*) the verifier evaluates at any point with few operations. We use

$$\mathbf{1}(w) = 1, \quad \mathbf{0}(w) = 0, \quad \text{id}(w) = w, \quad G_\eta(w) = \eta^w \quad (0^0 = 1),$$

with w read as an integer in $[0, N - 1]$ and then as an element of $\mathbb{F}$, for which $V_1(X) = \prod_{k < n} (1 + X^{2^k}) = V_1^*(X)$, $V_{G_\eta}(X) = V_1(\eta X)$, $V_{G_\eta^*}(X) = \prod_{k < n} (X^{2^k} + \eta^{2^k})$, and $V_{\text{id}} = T_n$ with $S_0 = 1$, $T_0 = 0$, $S_{k+1} = (1 + X^{2^k})S_k$, $T_{k+1} = (1 + X^{2^k})T_k + 2^k X^{2^k} S_k$; each costs $O(n)$ operations. An *affine form* is $\lambda = \sum_{i \in S} c_i w_i + \sum_j d_j P_j$ with a finite set S of commitments (its *support*, listed in the instance; a coefficient c_i may vanish), public tables P_j and coefficients $c_i, d_j \in \mathbb{F}$. The form is *public* if $S = \emptyset$. Its word at ξ is $\sum_i c_i w_i(\xi) + \sum_j d_j V_{P_j}(\xi)$, and for tables f_i of the w_i its table is $\sum_i c_i f_i + \sum_j d_j P_j$.

Proposition 25 (Π_{Batch} on affine forms). *Let the arguments of the statements of Section 4 be affine forms, the coefficients and the parameters of the public tables being part of the instance.*

- *Relation: tables are required for the committed words in the support of some argument, within fibre distance δ ; every statement holds for the tables of its affine forms. Public tables carry no condition.*
- *Curve: the committed words occurring directly, the reversals of those occurring reversed, every $w_Q^{(j)} = \text{ev}_L(V_\lambda(\gamma X))$ (public part included), every $w_A^{(j)}$ and h_j , and the quotient words $q[\lambda, y, x]$ of the forms λ that are not public. When the form of a quotient word is public, the verifier checks $y = V_\lambda(x)$ directly instead.*

Then completeness, Lemma 23 and Theorem 24 hold with $t + 1$ the number of words in the curve. The verifier's work per query point grows by $O(n)$ per public table.

Proof. On the agreement set T of Lemma 23, the word of an affine form agrees with the polynomial $\sum_i c_i P_{w_i} + \sum_j d_j V_{P_j}$ of degree $< N$, since public tables agree with their polynomials everywhere. The proofs of Lemma 23 and Theorem 24 use only such agreements, Lemmas 4 and 13, and the decodings of the committed words, which determine the tables of the affine forms; the case analysis of Theorem 18 uses only the polynomial V_λ of the decoded form. A direct check $y = V_\lambda(x)$ gives the conclusion of Lemma 13 for a public form. $\square$

5.2 The lincheck

Definition 26 (Index, relation). A sparse $\mathbf{M}_j \in \mathbb{F}^{N \times N}$ ($j < J$) with at most N nonzero entries is listed as $(r_{j,k}, c_{j,k}, v_{j,k})_{k < N}$, padded with $v_{j,k} = 0$, $r_{j,k} = c_{j,k} = 0$. The *index* of $(\mathbf{M}_j)_{j < J}$ consists of the commitments Enc^T of the tables $\text{row}_j(k) = r_{j,k}$, $\text{col}_j(k) = c_{j,k}$, $\text{val}_j(k) = v_{j,k}$ and of the *combined* multiplicities $m_R(w) = \#\{(j, k) : r_{j,k} = w\}$, $m_C(w) = \#\{(j, k) : c_{j,k} = w\}$, computed once by an honest indexer (soundness uses only that the $\text{row}_j, \text{col}_j, \text{val}_j$ and m_R, m_C are honest), all

in one Merkle tree. The lincheck relation holds for commitments $(w_z, (w_{a_j})_j)$ and tables $(z, (a_j)_j)$ if all are within fibre distance δ of Enc^T of the tables and $a_j = \mathbf{M}_j z$ for every j .

The reduction rests on $\sum_w \eta^w(\mathbf{M}_j z)(w) = \sum_k v_{j,k} \eta^{r_{j,k}} z(c_{j,k})$: the prover commits to $e_{j,k} = \eta^{r_{j,k}}$, $\zeta_{j,k} = z(c_{j,k})$ and $p_{j,k} = e_{j,k} \zeta_{j,k}$, and proves the JN lookups $(r_{j,k}, e_{j,k}) \in \{(w, \eta^w)\}$ with *one* LogUp, and the JN lookups $(c_{j,k}, \zeta_{j,k}) \in \{(w, z(w))\}$ with another.

Definition 27 (Π_{Lin}). Instance: the index of $(\mathbf{M}_j)_{j < J}$ and commitments $w_z, (w_{a_j})_j$.

1. The verifier sends η , uniform in $\mathbb{F}$. The prover commits to tables e_j, ζ_j, p_j (honestly $e_{j,k} = \eta^{r_{j,k}}$, $\zeta_{j,k} = z(c_{j,k})$, $p_{j,k} = e_{j,k} \zeta_{j,k}$).
2. The verifier sends x_R, y_R, x_C, y_C , uniform in $\mathbb{F}$. The prover commits to tables $\phi_j^R, \psi_j^R, \phi_j^C, \psi_j^C$ and sends $s_j \in \mathbb{F}$; honestly

$$\begin{aligned} \phi_{j,k}^R &= \frac{1}{x_R - r_{j,k} - y_R e_{j,k}}, & \psi_w^R &= \frac{m_R(w)}{x_R - w - y_R \eta^w}, \\ \phi_{j,k}^C &= \frac{1}{x_C - c_{j,k} - y_C \zeta_{j,k}}, & \psi_w^C &= \frac{m_C(w)}{x_C - w - y_C z(w)}, \end{aligned}$$

with $\psi_w = 0$ when $m(w) = 0$; the honest prover aborts if a denominator with $m(w) \neq 0$ vanishes.

3. The statements, proved by one Π_{Batch} on affine forms (Proposition 25) whose own challenges γ, θ, β are drawn after step 2: for every j ,

$$\begin{aligned} &\text{IP}(a_j, G_\eta, s_j), \quad \text{IP}(\text{val}_j, p_j, s_j), \quad \text{Had}(e_j, \zeta_j, p_j), \\ &\text{Had}(\phi_j^R, x_R \mathbf{1} - \text{row}_j - y_R e_j, \mathbf{1}), \quad \text{Had}(\phi_j^C, x_C \mathbf{1} - \text{col}_j - y_C \zeta_j, \mathbf{1}), \end{aligned}$$

and once

$$\begin{aligned} &\text{Had}(\psi^R, x_R \mathbf{1} - \text{id} - y_R G_\eta, m_R), \quad \text{IP}(\sum_j \phi_j^R - \psi^R, \mathbf{1}, 0), \\ &\text{Had}(\psi^C, x_C \mathbf{1} - \text{id} - y_C z, m_C), \quad \text{IP}(\sum_j \phi_j^C - \psi^C, \mathbf{1}, 0), \end{aligned}$$

that is $5J + 4$ statements.

The honest prover aborts with probability at most $2N/|\mathbb{F}|$ (a denominator $x - w - y \cdot$ with $m(w) \neq 0$ vanishes; the denominators of the ϕ are among them), and is otherwise accepted with the probability of Proposition 25, at least $1 - 3M/|\mathbb{F}|$.

Lemma 28 (LogUp with pairs). *Let pairs $(\alpha_k, \alpha'_k)_{k < K}$ and $(\omega_w, \omega'_w)_{w < N_t}$ in $\mathbb{F}^2$, $m_w \in \mathbb{F}$, and $\text{char } \mathbb{F} > K$. If some $(\alpha_{k_0}, \alpha'_{k_0})$ is not among the (ω_w, ω'_w) , then for x, y uniform in $\mathbb{F}$,*

$$\Pr \left[\sum_k \frac{1}{x - \alpha_k - y \alpha'_k} = \sum_w \frac{m_w}{x - \omega_w - y \omega'_w} \text{ with all denominators nonzero} \right] \leq \frac{K + 2N_t - 1}{|\mathbb{F}|}.$$

Proof. Fix y and write $c_k = \alpha_k + y \alpha'_k$, $t_w = \omega_w + y \omega'_w$. The equation $c_{k_0} = t_w$ is $\alpha_{k_0} - \omega_w = y(\omega'_w - \alpha'_{k_0})$; it has no solution y when $\alpha'_{k_0} = \omega'_w$ (then $\alpha_{k_0} \neq \omega_w$) and one otherwise, so c_{k_0} is a table value t_w for at most N_t values of y . For the other y , let $S = \{c_k\}$, $n_c = \#\{k : c_k = c\}$ and

$$\text{Num}(X) = \sum_{c \in S} n_c \prod_{c' \in S \setminus \{c\}} (X - c') \prod_w (X - t_w) - \sum_w m_w \prod_{c \in S} (X - c) \prod_{v \neq w} (X - t_v),$$

of degree $\leq K + N_t - 1$. At $X = c_{k_0}$ only the term of $c = c_{k_0}$ survives, and it is $n_{c_{k_0}} \in [1, K]$ (nonzero in $\mathbb{F}$) times nonzero products, so $\text{Num} \neq 0$. Where all denominators are nonzero, the equation multiplied by $\prod_{c \in S} (x - c) \prod_w (x - t_w)$ is $\text{Num}(x) = 0$, which holds for at most $K + N_t - 1$ values of x . $\square$

Theorem 29 (Lincheck reduction). *Suppose the index is honest, the decodings z, a_j of w_z, w_{a_j} exist and $a_{j_0} \neq \mathbf{M}_{j_0} z$ for some j_0 , and fix a deterministic prover. Let E be the event, over η, x_R, y_R, x_C, y_C , that every word committed by the prover in steps 1–2 has a decoding within fibre distance δ and that these decodings, with z, a_j and the index, satisfy all the statements of step 3. Then*

$$\Pr[E] \leq \varepsilon_{\text{Lin}} = \frac{N-1}{|\mathbb{F}|} + 2 \frac{N + (JN + 2N - 1)}{|\mathbb{F}|} = \frac{(2J+7)N-3}{|\mathbb{F}|}.$$

Proof. On E : from $\text{Had}(\phi_j^R, x_R \mathbf{1} - \text{row}_j - y_R e_j, \mathbf{1})$, every $x_R - r_{j,k} - y_R e_{j,k}$ is nonzero and $\phi_{j,k}^R$ is its inverse. From $\text{Had}(\psi^R, \dots, m_R)$, $\psi_w^R(x_R - w - y_R \eta^w) = m_R(w)$; unless $x_R = w + y_R \eta^w$ for some w (probability $\leq N/|\mathbb{F}|$ over x_R , for every y_R), $\psi_w^R = m_R(w)/(x_R - w - y_R \eta^w)$. The last row statement gives $\sum_{j,k} \phi_{j,k}^R = \sum_w \psi_w^R$. The decodings e_j are fixed before x_R, y_R , and $w \mapsto w \in \mathbb{F}$ is injective on $[0, N-1]$ since $\text{char } \mathbb{F} > N$; so by Lemma 28 with $K = JN$ and $N_t = N$, except with probability $(JN + 2N - 1)/|\mathbb{F}|$, every $(r_{j,k}, e_{j,k})$ is a pair (w, η^w) , that is $e_{j,k} = \eta^{r_{j,k}}$. In the same way, except with probability $(N + JN + 2N - 1)/|\mathbb{F}|$, $\zeta_{j,k} = z(c_{j,k})$ for all j, k (the decodings ζ_j and z are fixed before x_C, y_C). Then $\text{Had}(e_{j_0}, \zeta_{j_0}, p_{j_0})$ gives $p_{j_0,k} = \eta^{r_{j_0,k}} z(c_{j_0,k})$, and the first two statements give $\sum_w \eta^w a_{j_0}(w) = s_{j_0} = \sum_k v_{j_0,k} \eta^{r_{j_0,k}} z(c_{j_0,k}) = \sum_w \eta^w (\mathbf{M}_{j_0} z)(w)$. So η is a root of the nonzero polynomial $\sum_w (a_{j_0} - \mathbf{M}_{j_0} z)(w) X^w$ of degree $\leq N-1$, fixed before η : probability $\leq (N-1)/|\mathbb{F}|$. $\square$

For $J = 1$, $\varepsilon_{\text{Lin}} = (9N - 3)/|\mathbb{F}|$. Separate lookups for each matrix would cost $J(4N - 1)/|\mathbb{F}|$ per kind of lookup instead of $((J + 3)N - 1)/|\mathbb{F}|$, and $J - 1$ more words ψ and more statements per kind.

Theorem 30 (Soundness of the lincheck). *If the index is honest and $(w_z, (w_{a_j})_j)$ has no witness for the lincheck relation, every prover of Π_{Lin} followed by Π_{Batch} is accepted with probability at most $\varepsilon_{\text{Lin}} + \varepsilon_{\text{Batch}}$, with $\varepsilon_{\text{Batch}}$ of Theorem 24 for the $t + 1$ words of step 3.*

Proof. It suffices to consider deterministic provers. Let τ be the transcript of steps 1–2; the batch instance $I(\tau)$ and the event E of Theorem 29 are functions of τ , and given τ the rest of the prover is a prover of Π_{Batch} facing fresh challenges. If w_z or some w_{a_j} has no decoding, $I(\tau)$ has no witness for every τ . Otherwise $a_{j_0} \neq \mathbf{M}_{j_0} z$ for some j_0 , and for $\tau \notin E$, $I(\tau)$ has no witness: by unique decoding a witness would consist of the decodings, which either do not exist or violate a statement. In both cases Theorem 24 gives $\Pr[\text{accept} \mid \tau] \leq \varepsilon_{\text{Batch}}$ for $\tau \notin E$, so the acceptance probability is at most $\Pr[E] + \varepsilon_{\text{Batch}}$. $\square$

5.3 R1CS

Definition 31 (R1CS relation). Public: sparse $A, B, C \in \mathbb{F}^{N \times N}$ (with an honest index), input positions $I \subseteq [0, N-1]$ and an input $x \in \mathbb{F}^I$. Let $\hat{x}$ be the table equal to x on I and 0 elsewhere, and χ the 0/1 table of I (both public tables; for $I = [0, |I| - 1]$, $V_\chi(X) = (X^{|I|} - 1)/(X - 1)$, and in general $V_{\hat{x}}, V_\chi$ cost $O(|I|)$). The relation holds for commitments (w_w, w_a, w_b, w_c) and tables (w, a, b, c) if all four are within fibre distance δ , w vanishes on I , and with $z = \hat{x} + w$: $a = Az$, $b = Bz$, $c = Cz$ and $a \circ b = c$.

Theorem 32 (R1CS without sumcheck). *Run Π_{Lin} for $(\mathbf{M}_0, \mathbf{M}_1, \mathbf{M}_2) = (A, B, C)$ and $(a_0, a_1, a_2) = (a, b, c)$ (so $J = 3$ and $\text{char } \mathbb{F} > 3N$) on the affine form $z = \hat{x} + w_w$, and prove its 19 statements, $\text{Had}(a, b, c)$ and $\text{Had}(w, \chi, \mathbf{0})$ in one Π_{Batch} on affine forms. If the instance has no witness, every prover is accepted with probability at most*

$$\varepsilon_{\text{Lin}} + \varepsilon_{\text{Batch}} \leq \frac{15N - 5 + tM}{|\mathbb{F}|} + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa,$$

with $t + 1 = 127$ words: 23 read directly, 17 reversed, 13 words w_Q , 21 words w_A , 21 words h and 32 quotient words, for every instance and every value of the challenges.

Proof. As in Theorem 30, with one false statement: if some decoding is missing, the batch instance never has a witness; if $a \neq Az$, $b \neq Bz$ or $c \neq Cz$ for the decodings, only the event E of the lincheck matters, of probability $\leq \varepsilon_{\text{Lin}} = (13N - 3)/|\mathbb{F}|$; otherwise $a \circ b \neq c$ or w does not vanish on I , and $\text{Had}(a, b, c)$ or $\text{Had}(w, \chi, \mathbf{0})$ is false for every value of the challenges. $\square$

Completeness. If $w_w = \text{Enc}^T(w)$ and $w_{a_j} = \text{Enc}^T(a_j)$ for a witness, the honest prover is accepted with probability at least $1 - 2N/|\mathbb{F}| - 3M/|\mathbb{F}|$: outside the challenges at which some $x_R - w - y_R \eta^w$ or $x_C - w - y_C z(w)$ vanishes (probability $\leq 2N/|\mathbb{F}|$), the honest tables satisfy the 21 statements (the two sum statements by $\sum_{j,k} f(r_{j,k}) = \sum_w m_R(w) f(w)$ and its column analogue), and Proposition 25 applies.

Every step is a coefficient extraction, a lookup by logarithmic derivatives, or a Hadamard check: an R1CS argument on Kronecker-FRI commitments needs no sumcheck.

Status. The algebra of Π_{Lin} is checked in the clear by the Rust module `lincheck` (closed forms of V_1 , V_{id} , V_{G_η} ; honest runs with padded matrices; a wrong a caught by the weighted sums; a forged ζ_k or e_k that repairs the weighted sums caught by the column or row lookup). The compiled protocol is implemented: the module `affine` runs Π_{Batch} on affine forms (Proposition 25) with committed words grouped in Merkle trees whose leaves hold one value of each word of a group, and the module `r1cs` implements the indexer, prover and verifier of Theorem 32: one index tree of 11 words ($\text{row}_j, \text{col}_j, \text{val}_j, m_R, m_C$), combined row and column lookups, 21 statements and $t + 1 = 127$ words (the words of a form are its declared support, so the count does not depend on the challenges; test `word_count_127`, the count of the Lean theorem `tA_r1stmts`); the proof opens six trees per query (witness, index, the two lincheck groups, the two groups of Π_{Batch}). Its tests accept honest proofs and reject a wrong public input, an unsatisfied constraint, a witness that is nonzero on an input position, a false lincheck alone, and a false lincheck repaired by a forged ζ_k or e_k ; removing the sum statement of the column (row) lookup makes the forged ζ_k (e_k) accepted, so each lookup is exercised. The Lean formalisation of this section is described in Section 7.

n	constraints	index (ms)	prove (ms)	verify (ms)	proof (KiB)
10	512	3.1	59.6	20.8	890.0
12	2048	12.1	272.0	39.2	1701.1
14	8192	54.1	1223.0	43.4	1843.8
16	32768	259.3	5469.1	43.6	1997.1

Table 2: The R1CS prototype on the instances of `r1cs::sample_instance` ($N/2$ constraints $(Az)_i(Bz)_i = z_{N/2+i}$, two nonzero entries per row of A and B , two public inputs); $t + 1 = 127$ words; configuration KF-8 ($\ell = \max(n - 8, 1)$), 148 queries, 32-byte salts, $\mathbb{F}_{p^2}$; one core; medians. Source: `bench/r1cs.csv`, `cargo run --release --example r1cs_bench`.

Compared with separate lookups for each matrix and three index trees (version 8 of this note: 179 words, eight trees), at $n = 16$ the prover is 30% faster ($7802 \rightarrow 5469$ ms), the verifier 36% faster ($67.8 \rightarrow 43.6$ ms) and the proof 26% smaller ($2694 \rightarrow 1997$ KiB).

Comparison with a sumcheck-based argument. The Rust module `spartan` implements the core of Spartan [5] on the same commitment, field, hash, configuration and instances: the prover commits the witness table w only; an outer sumcheck of degree 3 proves $\sum_x \text{eq}(\tau, x) (\hat{A}z(x) \hat{B}z(x) -$

$\tilde{C}z(x) = 0$, an inner sumcheck of degree 2 reduces the three claims $\tilde{A}z(r_x), \tilde{B}z(r_x), \tilde{C}z(r_x)$ to one evaluation of $\tilde{z}$, and one Π_{KF} opening of w at r_y ends the proof; the verifier evaluates $\tilde{\mathbf{M}}(r_x, r_y)$ itself, in time linear in the number of nonzero entries. Its tests accept honest proofs and reject a wrong public input, an unsatisfied constraint, a wrong claim, a wrong opening value and a tampered round polynomial.

n	this note (Theorem 32)			Spartan core		
	prove (ms)	verify (ms)	proof (KiB)	prove (ms)	verify (ms)	proof (KiB)
10	59.6	20.8	890.0	3.4	1.4	67.8
12	272.0	39.2	1701.1	14.7	2.7	135.6
14	1223.0	43.4	1843.8	60.8	4.2	182.3
16	5469.1	43.6	1997.1	288.1	11.7	251.1
18				1163.8	78.0	327.0
20				5269.2	416.8	390.1

Table 3: The R1CS argument of this note and the Spartan core on the instances of `r1cs::sample_instance`; KF-8, 148 queries, 32-byte salts, $\mathbb{F}_{p^2}$; one core, same session; medians. The index of this note (259 ms at $n = 16$) is computed once and not included. Source: `bench/r1cs.csv`, `bench/spartan.csv`, `cargo run --release --example spartan_bench`.

The measurements say:

- *Prover.* The Spartan core is 17–20 times faster (288 against 5469 ms at $n = 16$). A profile of our prover at $n = 16$: witness group 0.12 s, round-1 lincheck words 0.4 s, round-2 words 0.35 s, and 4.4 s in Π_{Batch} on affine forms, of which 0.95 s for the 21 products $U_j K_j$, 0.8 s for committing the 21 words w_A , 1.06 s for the virtual words h_j , 0.57 s for the batched word, 0.47 s for the quotient words, 0.53 s for the words w_Q and 0.9 s for the openings; the folding test itself takes 0.02 s. The cost is spread over the statements: each costs a product of two polynomials of degree $< N$ over $\mathbb{F}_{p^2}$ and words of length $M = 4N$, where a sumcheck round costs a linear pass over tables of length N .
- *Baseline.* The measured core is a lower bound for Spartan’s prover: full Spartan adds an argument that commits the matrices once and proves the value $\tilde{\mathbf{M}}(r_x, r_y)$, which the core leaves to the verifier.
- *Proof size.* Ours is 8 times larger at $n = 16$ (1997 against 251 KiB): six trees are opened per query instead of the two trees of one Π_{KF} opening.
- *Verifier.* Ours is succinct: 44 ms from $n = 14$ to 16, with the index committed once. The verifier of the Spartan core is linear in the number of nonzero entries: 11.7, 78 and 417 ms at $n = 16, 18, 20$.

6 Kronecker-FRI in sumcheck-based arguments: the two routes measured

The protocols of Sections 2–5 are one of two ways to use Kronecker-FRI for statements on committed tables; the other is a sumcheck, which reduces the statement to the values of the tables at one random point, followed by an opening. This section measures both on the same commitment.

Post-quantum setting. Kronecker-FRI is hash-based: its security rests on the hash function and on the proximity gaps of Reed–Solomon codes, with no discrete-logarithm or pairing assumption. Every argument built on it is post-quantum in this sense, the sumcheck route included, since a sumcheck is an information-theoretic reduction. The sumcheck-free protocols are therefore an

alternative to the sumcheck with the same security basis; they change the cost, the number of rounds and the verifier, not the security assumptions.

One statement. The Rust module `sumcheck` proves one inner product and one Hadamard check by both routes with the same commitments (one tree per table) and the same proof engine (`affine`): the *sumcheck-free* route proves $\text{IP}(a, b, S)$ or $\text{Had}(a, b, c)$; the *sumcheck* route runs a sumcheck of degree 2 for the inner product, or the zerocheck $\sum_x \text{eq}(\tau, x)(a(x)b(x) - c(x)) = 0$ of degree 3 for the Hadamard check, and then proves *one* evaluation statement on the affine form $\sum_i \rho^i w_i$ at the sumcheck point, a batched opening. Only the reduction differs. Tests accept honest proofs of both routes and reject a wrong sum, a false Hadamard instance and wrong values at the sumcheck point.

n	inner product						Hadamard check					
	sumcheck-free			sumcheck			sumcheck-free			zerocheck		
	P	V	KiB	P	V	KiB	P	V	KiB	P	V	KiB
12	11.9	3.2	178	12.6	3.7	179	22.9	5.4	268	13.2	4.0	223
14	55.0	4.0	233	55.5	4.5	235	101.4	6.3	359	58.5	4.7	286
16	225.2	4.6	310	227.0	5.2	310	431.7	7.2	449	257.4	5.4	368
18	901.9	5.4	396	937.1	6.0	395	1746.1	7.8	557	1023.5	6.4	473
20	4197.4	6.0	468	4391.9	7.1	468	8131.0	9.2	648	4695.4	7.8	556

Table 4: Prove time P (ms), verify time V (ms) and proof size of one statement by the two routes, on the same commitments and engine; commitments excluded; KF-8, 148 queries, 32-byte salts, $\mathbb{F}_{p^2}$; one core; medians. Source: `bench/primitives.csv`, `cargo run --release --example primitives_bench`. The dedicated module `ip` proves an inner product in 109 ms at $n = 16$ (Table 5); the engine is generic, and the comparison uses it for both routes.

- *Inner product.* The two routes cost the same: 225 against 227 ms at $n = 16$, equal proof sizes, a verifier 0.6 ms faster without sumcheck. Both prove one opening polynomial and one virtual word; the sumcheck adds a linear pass per round and n rounds of interaction.
- *Hadamard check.* The zerocheck is 1.7 times faster (257 against 432 ms at $n = 16$) with a 18% smaller proof: the sumcheck-free check adds the word w_Q over $\mathbb{F}_{p^2}$ and three quotient words, the zerocheck one linear pass per round.
- *R1CS* (Section 5). The gap grows to 19 times against the Spartan core, because the lookups multiply the statements over $\mathbb{F}_{p^2}$.
- *Kronecker-FRI inside Spartan.* In the Spartan core at $n = 16$, the commitment takes 73.5 ms, the two sumchecks 29.2 ms and the opening 178.3 ms (62%); at $n = 20$: 1372, 804 and 3035 ms. The cost of a sumcheck-based argument on Kronecker-FRI is the cost of its commitment and opening.
- *Rounds and verifier.* The sumcheck-free route has no rounds that grow with n beyond the folding test, which both routes share; the sumcheck route adds n rounds, n challenges and n round polynomials. The R1CS verifier of Section 5 stays at 44 ms with a committed index, where the verifier of the Spartan core is linear in the number of nonzero entries.

7 Relation to known techniques and status

Writing an inner product as the middle coefficient of a product with a reversed polynomial is classical, and appears in pairing-based multilinear commitments (Mercury [3] uses the table form with KZG). DEEP quotients for out-of-domain values are standard in FRI-based STARKs. The new points here are: (1) the reversed virtual word w_b^* , which reads the second vector from the

same Reed–Solomon commitment thanks to $L^{-1} = L$; (2) the reuse of Kronecker-FRI’s opening polynomial, virtual word and single folding test, with a fully proved soundness bound (Lemma 9); (3) the table-form evaluation kernel E_z^* , which gives evaluation openings on the same commitments; (4) a Hadamard check in which the out-of-domain values are DEEP quotient words inside the same batch as the opening polynomial, so that one degree-8 curve and one folding test carry the whole argument (Lemma 17).

Prover cost and the sumcheck baseline. The prover of Π_{IP} computes one product $V_a V_b^*$ of two polynomials of degree $< N$ over the base field (three NTTs of size $2N$), one NTT of size M and one Merkle tree for w_A , and then runs the folding test; every round-1 object lies in the base field, since V_a, V_b do. The natural baseline is the sumcheck route: n rounds of a degree-2 sumcheck on $\sum_w a(w)b(w)$, followed by openings of a and b at the random point. For that baseline the fastest prover we know of in our own work is the tree circuit of [6], which produces the evaluations, hypercube sums and sumcheck round polynomials of a multilinear polynomial from one binary-tree recursion; Section 6 compares the two routes on the same commitment and engine.

Status.

- *Lean 4.* Sections 1–2 are formalised in the Kronecker-FRI Lean project, files `TableForm.lean` and `InnerProduct.lean`, without `sorry` and with the single axiom of that project (correlated agreement for curves, KF Thm 2.16, here with $t = 3$). Lemma 2: `eval_table`; Lemma 3: `inner_product`; Lemma 4: `splitG_iff`, `natDegree_PhiG_le`, `identityG`; Lemma 5: `wrev_ev`, `wrev_eq_iff`, `inv_image_closed`, `ncard_inv_image`; completeness: `honestIP_prob_one`; Lemma 9: `batching_IP`; soundness: `soundness_IP`, and `soundnessC_IP` for any folding arity 2^k ; knowledge and binding: `knowledge_IP`, `binding_IP`. The common step of KF Lemma 7.4 and Lemma 9, correlated agreement on one fibre-closed set for a curve of any degree, is the separate theorem `curve_agreement`.
- *Rust.* The module `ip` of the Kronecker-FRI crate implements Π_{IP} with Merkle caps, arity 2^k and Fiat–Shamir, on the commitments of `pcs`. Its tests check Lemmas 3, 4 and 5(1), completeness for $n \leq 10$, two rates and arities 2 to 16, and the rejection of three cheating provers (a wrong S ; a wrong S absorbed into a polynomial A of degree N , as in KF Remark 4.13; opening polynomials computed from a table $b' \neq b$).
- *Lean 4, Section 3.* File `Hadamard.lean`, same axiom, no `sorry`. Lemma 13: `deep_step`, `deep_honest`; completeness: `honestHad_accepts`, and `honestHad_prob` for the bound $1 - 3M/|\mathbb{F}|$; Lemma 17: `batching_Had`; Theorem 18: `prob_bad_le` (the bad pairs (γ, θ)), `soundness_Had`, and `soundnessC_Had` for any folding arity 2^k ; knowledge: `knowledge_Had`. The step “for good (γ, θ) , the argument of KF Thm 7.8” is the separate theorem `ft_family_bound`: for any family of words $w_0(\beta)$, the acceptance probability is at most $|\mathcal{G}|/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$.
- *Lean 4, Section 4.* File `BatchStmts.lean`, same axiom, no `sorry`. Statements: `BStmt` (`eval`, `ip`, `had`), `Dset`, `Bset`, `RelBatchS`; the words are indexed by the finite type `BWord` ($D \oplus B \oplus \mathcal{H} \oplus [J] \oplus [J] \oplus \mathcal{H} \times [3]$) with a fixed enumeration, and `curve_agreement_idx` is correlated agreement for such an index type. Lemma 23: `batching_B`; Theorem 24: `prob_bad_had` (one false Hadamard check), `prob_bad_B`, `soundness_B`, `soundnessC_B` (any folding arity); knowledge: `knowledge_B`; completeness: `bwords_honest`, `honestB_accepts`, `honestB_prob`.
- *Lean 4, Section 5.* Files `AffineBatch.lean`, `Lincheck.lean` and `R1CS.lean`, no `sorry`, same axiom. Proposition 25: forms `AForm` (declared support, coefficients, public table) and statements `AStmt` on them, the relation `RelBatchA`, the direct checks `DirOK`; Lemma 23 on forms:

`batching_A`; soundness and knowledge: `prob_bad_A`, `soundness_A`, `knowledge_A`; completeness: `awords_honest`, `honestA_accepts`, `honestA_prob`. Lemma 28: `logup_prob`, for any finite index types of the K lookups and N_t table pairs. Theorem 29: `lincheck_reduction`, for J matrices with combined lookups, on tables indexed by any finite set with a numbering (`Numbering`; Fin N , or the hypercube with $w \mapsto \sum_k w_k 2^k$), for any deterministic prover given by its tables as functions of the challenges; one failing lookup: `prob_lookupFail`; the two-phase bound `prob_two_phase` and its instances `lincheck_sound`, `rics_sound`. Theorem 32 end to end: `rics_soundness`, for the protocol with the honest index, the prover’s words as functions of η, x_R, y_R, x_C, y_C and a prover of Π_{Batch} for each value of them, with the bound $(13N - 3)/|\mathbb{F}| + 2(N - 1)/|\mathbb{F}| + 126M/|\mathbb{F}| + \varepsilon_{\text{fold}} + (1 - \delta)^\kappa$; the 21 statements `r1stmts`, the count $t = 126$ (`tA_r1stmts`), and the step from a batch witness to the lincheck statements on the decodings (`r1_bridge`). Completeness of the R1CS argument: `honestR1`, `logup_honest`, `honest_holds` (the 21 statements hold for the honest tables when no denominator vanishes), `rics_completeness` (probability $\geq 1 - 2N/|\mathbb{F}| - 3M/|\mathbb{F}|$). The closed forms of the public tables, file `PublicTables.lean`: $V_1 = \prod_k (1 + X^{2^k})$ (`kronEnc_one`), $V_{G_\eta}(x) = V_1(\eta x)$ (`eval_kronEnc_geo`), $V_{\text{id}} = XV_1'$ (`kronEnc_id`), and the recursion S_k, T_k used by the verifier (`kronEnc_one_eq_recS`, `kronEnc_id_eq_recT`).

- *Rust, Section 4.* The module `batch` implements Π_{Batch} with one multi-value Merkle tree for all w_Q and one for all w_A ; its tests check Lemma 2 for E_z^* , completeness for mixed batches (two Hadamard checks sharing a table, an inner product, two evaluations) for $n \leq 10$ and arities 2 to 16, and the rejection of a batch with one false evaluation, one false inner product, or one false Hadamard check (also against a prover that sets y_3 to the weighted inner product, which only the quotient word $q[w_c, y_3, \gamma]$ catches).
- *Rust, Section 3.* The module `had` implements Π_{Had} on the shared folding-test code of module `ft`, drawing γ, θ by rejection sampling outside the base field $\mathbb{F}_p \supseteq L$, so that no pole lies in L and completeness is perfect (the two first terms of ε_{Had} become $(N - 1)/(|\mathbb{F}| - p)$ each); its tests check completeness for $n \leq 10$, two rates and arities 2 to 16, and the rejection of a wrong y_1 , of a wrong Q (with $y_1 = Q(\theta)$), and of a false instance (c differing from $a \circ b$ in one entry) with three choices of y_3 . Removing $q[w_c, y_3, \gamma]$ from the batch makes the false instance accepted, and a wrong pole in the verifier breaks completeness, so the tests exercise each part.

n	prove (ms)			verify (ms)			proof (KiB)		
	KF	IP	Had	KF	IP	Had	KF	IP	Had
12	7.5	6.2	17.6	2.09	2.13	4.31	138.1	171.4	289.1
14	32.2	25.9	74.0	2.56	2.60	4.97	182.3	233.4	371.1
16	141.4	109.2	314.7	3.24	3.25	5.63	249.8	308.1	458.8
18	548.2	437.8	1353.6	3.87	4.01	6.40	325.0	395.3	565.7
20	2736.6	1947.4	6004.5	4.59	5.13	7.39	391.7	471.1	662.0

Table 5: KF: one evaluation opening (Π_{KF}); IP: one inner product (Π_{IP}); Had: one Hadamard check (Π_{Had}). Configuration KF-8 of the Kronecker-FRI paper (rate 1/4, arity 8, caps of 128 nodes, $\ell = n - 8$), 148 queries, 32-byte salts, $\mathbb{F}_{p^2}$ over Goldilocks; one core of an Intel Xeon at 2.10 GHz; medians; commitments not included. Source: `bench/ip.csv`, `cargo run --release --example ip_bench`.

An inner product of two committed tables costs less prover time than one evaluation opening (the round-1 work is in the base field), about the same verifier time, and about 20% more proof, the extra openings of w_b on the inverse cosets. A Hadamard check costs about 2.2 evaluation openings of prover time (its polynomials Q, A, H lie in the extension field) and 1.7 times the proof, since

five oracles are opened per query instead of two; when a, b, c are committed together, one tree with three values per leaf removes two of the five authentication paths per query.

Next. On Kronecker-FRI, the cost of a sumcheck-based argument is the cost of its commitment and opening (Section 6); a faster opening therefore benefits both routes. For the sumcheck-free route, the cost to reduce is that of the lookups of Section 5.

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